Course Outline

Sem		Course Name	Th	Tut	Pra	Hrs	Cr
Sem I	HSC	Humanities – 1	2	0	0	2	2
	BSC	Mathematics – 1	3	1	0	4	4
	BSC	Physics	3	0	0	3	3
	BSC	Physics – Laboratory	0	0	2	2	1
	BSC	Environment Science	2	0	0	2	2
	ESC	Workshop Practices	0	0	2	2	1
	BSC	Biology for Engineers	2	0	0	2	2
	ESC	Computer Programming -I	1	0	0	1	1
	ESC	Computer Programming -I Laboratory	0	0	2	2	1
	HSC	Universal Human Values	1	0	0	1	1
	HSC	Indian Knowledge System	2	0	0	2	2
			16	**1**	**5**	**23**	**20**
Sem II	HSC	Humanities – 2	2	0	0	2	2
	BSC	Mathematics – 2	3	1	0	4	4
	BSC	Chemistry	3	0	0	3	3
	BSC	Chemistry – Laboratory	0	0	2	2	1
	ESC	Introduction to Electrical Engineering	3	0	0	3	3
	ESC	Introduction to Electrical Engineering-Laboratory	0	0	2	2	1
	ESC	Engineering Graphics	0	0	4	4	2
	HSC	Yoga, Health & Hygiene OR NCC/NSS	0	0	2	2	1
	HSC	Organizational Behaviour	1	0	0	1	1
	ESC	Computer Programming -II	1	0	0	1	1
	ESC	Computer Programming -II Laboratory	0	0	2	2	1
			13	**1**	**12**	**26**	**20**
Sem III	Pro	Civic and Social Service Internship	0	0	0	0	1
	ESC	Introduction to Artificial Intelligence (Department Specific)	3	0	0	3	3
	PC	Digital Circuits	3	0	0	3	3
	PC	Digital Circuits Laboratory	0	0	2	2	1
	PC	Data Structures and Algorithms	3	0	0	3	3
	PC	Data Structures and Algorithms Laboratory	0	0	2	2	1
	PC	Fundamentals of ICT	2	0	0	2	2
	PC	Electronics Devices and Circuits	2	0	0	2	2
	BSC	Discrete Mathematical Structures	3	1	0	4	4
			16	**1**	**4**	**21**	**20**
Sem IV	ESC	Industry 4.0 (Department Specific)	2	0	0	2	2
	ESC	Industry 4.0 – Laboratory (Department Specific)	0	0	2	2	1
	OE	Open Elective 1 (From Other School)	3	0	0	3	3
	PC	Database Management Systems	3	0	0	3	3
	PC	Database Management Systems Laboratory	0	0	2	2	1
	PC	Principles of Programming Languages	3	0	0	3	3
	PC	Principles of Programming Languages Laboratory	0	0	2	2	1

1st Semester

UG_1_T(CSE/ICT/EC/EE)					Computer Programming-I					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
1	0	0	1	1	25	50	25	-	-	100

COURSE OBJECTIVES

- To understand the usage of operators and data types.
- To apply different types of Conditional and looping statement.
- To create different types of data collections.
- To implement user defined function.
- To perform different operations upon files.

UNIT 1: Basics of Programming Introduction to Computer Programming, Features of C language, Structure of C program, program execution flow, C Tokens, variables, Data types, Operators, Decision control statements-if, switch, go to statement. Loop control structures- while, do-while, for loop, Break statement, Continue statement	**3 Hrs.**
UNIT 2: Derived Data types Array: one dimensional and multidimensional array, Declaration, initialization, Array Manipulations. Matrix operations, String-Basic Concepts, Inbuilt String manipulation Functions, Pointer, Pointer arithmetic, Pointer to pointer, Array of Pointers, Dynamic memory allocation.	**4 Hrs.**
UNIT 3: Function and Structure Introduction to user defined functions, Types of Functions, Call by value-call by reference, recursion, pointers to functions, Structures, Self-referential structures, typedef, Array of Structure, Union	**4 Hrs.**
UNIT 4: Basics of Files and Data Structures File handling in C, Different types of files, Operations on Files, File handling functions. Data Structure, definition, Types of data structures, Applications of data structures	**3 Hrs.**
	14 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Understand basics of programming.
CO2 - Identify different programming constructs for a problem.
CO3 - Apply appropriate derived data type for data storage.
CO4 - Prepare a user defined data type based on data attributes.
CO5 - Construct user defined functions for problem solving.
CO6 - Analyse different data structure based on application requirement.

TEXT/REFERENCE BOOKS

1. Kernighan & Ritchie, C Programming Language", PHI
2. K. N. King , "C Programming: A Modern Approach" , W.W. Nortorn
3. David Griffiths and Dawn Griffiths, "Head First C: A Brain-Friendly Guide", O'Reilly
4. E.Balaguruswamy, Programming in ANSI C, McGraw-Hill
5. Y.P. Kanetkar ,"Let us C", BPB Publication
6. Y.P. Kanetkar , "Pointers in C", BPB Publications

UG_1_P(CSE/ICT/EC/EE)					Computer Programming – I Laboratory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	0	2	1	2	--	--	--	25	25	50

COURSE OBJECTIVES

- To implement basic programming concepts.
- To create different types of data collections.
- To implement user defined function.
- To perform different file handling operations.

LIST OF EXPERIMENT

1.	**Introduction to Computer Programming**: Getting acquainted with Linux Terminal and basic commands, understanding compilation process through a simple C program, program execution flow, C Tokens, variables and keywords and identifiers, types of C constants and variables
2.	**Simple and formatted Input Output Operations:** Data types, Operators, Input /output statements in C, Formatted I/O, format specifiers, escaper sequences
3.	**Decision making and branching:** if, if-else, if-else ladder, switch, go to statement, conditional operator statement
4.	**Looping control structures:** while, do-while, for loop, Break statement, Continue statement
5.	**Basics of modular programming:** Introduction to modular programming through basic function syntax, introduction to header file creation, difference between #include <filename> and #include "filename"
6.	**Derived Data Type: Array and Strings:** One dimensional and multidimensional array, Declaration, initialization, Array Manipulations. Matrix operations, Basic Concepts, Inbuilt String manipulation with and without using inbuilt functions.
7.	**Derived Data Type: Structure and Union:** structure, arrays and structures, structures and functions, pointer to structure, self-referential structures, typedef, unions
8.	**Functions:** Introduction to user defined functions, Types of Functions, Call by value-call by reference, static and external variables, header file creation, recursion, pointers to functions, arrays and functions
9.	**Pointers:** Pointers basics, use of &, * operator in context to pointers, Pointer arithmetic, Array and String processing using pointer, pointer to pointer, Array of Pointers
10.	**File Handling in C:** File handling in C, Different types of files, Operations on Files, File handling functions.

COURSE OUTCOMES

On completion of the course, student will be able to

CO1 - Identify the use of appropriate naming conventions and programming style including appropriate comment density.
CO2 - Implement a basic C program using appropriate control structure.
CO3 - Apply appropriate derived data types based on data attributes.
CO4 - Develop the problem solution modularly.
CO5 - Design user defined functions for problem solving and reuse them across different programs.
CO6 - Apply suitable file handling and data structures.

TEXT/REFERENCE BOOKS

1. Kernighan & Ritchie, C Programming Language", PHI
2. K. N. King , "C Programming: A Modern Approach", W.W. Norton
3. E. Balaguruswamy, Programming in ANSI C, McGraw-Hill
4. Y.P. Kanetkar,"Let us C", BPB Publication
5. Y.P. Kanetkar, "Pointers in C", BPB Publication